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GenJSLU: A Unified Generative Framework for Joint Spoken Language Understanding with Heterogeneous Data

(/pdf?id=P0qnY1fFe3)

ACL ARR 2025 February Submission3553 Authors

15 Feb 2025 (modified: 23 Mar 2025) ACL ARR 2025 February Submission February, Senior Area Chairs, Area Chairs, Reviewers, Authors Revisions (/revisions?id=P0qnY1fFe3) CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/)

Abstract:
Spoken Language Understanding (SpokenSLU) involves multiple tasks, including Automatic Speech Recognition (ASR), Named Entity Recognition (NER), and Sentiment Analysis (SA). Traditional approaches either \train separate models for each task, limiting cross-task interaction and requiring task-specific architectures, or rely on fully annotated datasets, which are often scarce. In this work, we propose GenJSLU, a unified generative framework that addresses these challenges through two key innovations: a unified data representation that enables effective utilization of heterogeneous datasets, reducing reliance on fully annotated data, and a unified model architecture that enhances task interaction by formulating ASR, NER, and SA as a single generative task. Furthermore, GenJSLU is highly scalable and can be extended to large language models (LLMs), further improving SLU performance. Extensive experiments on public datasets demonstrate the effectiveness of our approach, achieving significant improvements over existing methods and highlighting its potential for future research.

Paper Type: Long
Research Area: Speech Recognition, Text-to-Speech and Spoken Language Understanding
Research Area Keywords: automatic speech recognition;spoken language grounding
Contribution Types: Model analysis & interpretability, NLP engineering experiment
Languages Studied: English
Reassignment Request Area Chair: This is not a resubmission
Reassignment Request Reviewers: This is not a resubmission
A1 Limitations Section: This paper has a limitations section.
A2 Potential Risks: No
A2 Elaboration: Our work has no potential risks.
B Use Or Create Scientific Artifacts: Yes
B1 Cite Creators Of Artifacts: Yes
B1 Elaboration: Section 4
B2 Discuss The License For Artifacts: Yes
B2 Elaboration: Section 4
B3 Artifact Use Consistent With Intended Use: Yes
B3 Elaboration: Section 4
B4 Data Contains Personally Identifying Info Or Offensive Content: Yes
B4 Elaboration: Section 4
B5 Documentation Of Artifacts: Yes
B5 Elaboration: Section 4
B6 Statistics For Data: Yes
B6 Elaboration: Section 4
C Computational Experiments: Yes
C1 Model Size And Budget: Yes
C1 Elaboration: Section 4
C2 Experimental Setup And Hyperparameters: Yes
C2 Elaboration: Section 4
C3 Descriptive Statistics: Yes
C3 Elaboration: Section 4

C4 Parameters For Packages: Yes
C4 Elaboration: Section 4
D Human Subjects Including Annotators: No
D1 Instructions Given To Participants: No
D1 Elaboration: No participants used.
D2 Recruitment And Payment: No
D2 Elaboration: No participants used.
D3 Data Consent: No
D3 Elaboration: The data is public.
D4 Ethics Review Board Approval: N/A
D4 Elaboration: The data is public.
D5 Characteristics Of Annotators: No
D5 Elaboration: No participants used.
E Ai Assistants In Research Or Writing: No
E1 Information About Use Of Ai Assistants: No
E1 Elaboration: No Ai used.
Author Submission Checklist: yes
Submission Number: 3553

Discussion (/forum?id=P0qnY1fFe3#discussion)

Committee Members Chat (/forum?id=P0qnY1fFe3#committee-chat)

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10 / 10 replies shown

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Official Review of Submission3553 by Reviewer RrUc

Edit

Official Review by Reviewer RrUc (Peijie Huang (/profile?id=~Peijie_Huang1))
23 Mar 2025, 22:52 (modified: 28 Mar 2025, 17:27)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer RrUc
Revisions (/revisions?id=rN2xmFk1Xd)

Paper Summary:
The paper proposes GenJSLU, a unified generative framework for joint Spoken Language Understanding (SLU) tasks, including Automatic Speech Recognition (ASR), Named Entity Recognition (NER), and Sentiment Analysis (SA). Instead of training separate models for each task, GenJSLU integrates them into a single generative model, enhancing task interaction and leveraging heterogeneous datasets. The framework reformulates these tasks under a unified representation and generative decoding process, improving overall SLU performance. Experimental results on SLUE-VoxPopuli and SLUE-VoxCeleb datasets demonstrate its superiority over baseline models.

Summary Of Strengths:
1. Unified SLU Modeling: Merges ASR, NER, and SA tasks into a single framework, reducing redundancy and improving performance.
2. Enhanced Task Interaction: Enables richer cross-task information sharing, boosting overall understanding.
3. Improved Data Utilization: Leverages heterogeneous datasets more effectively than traditional approaches.
4. Strong Empirical Results: Outperforms previous models on SLUE-VoxPopuli and SLUE-VoxCeleb datasets.

Summary Of Weaknesses:
1. The model size used in the experiment is too small. The authors should evaluate at least a 7B model.

2. Isn't NER in SLU referred to as slot filling?
3. The selection of baselines is not convincing. There are many joint baselines, and the simplest approach would be training all tasks jointly with different weights using a multi-model setup.
4. The paper lacks novelty, as it merely combines multiple tasks into a framework, which is a common practice in NLU.

Comments Suggestions And Typos:
See weaknesses section.

Confidence: 4 = Quite sure. I tried to check the important points carefully. It's unlikely, though conceivable, that I missed something that should affect my ratings.
Soundness: 2.5
Excitement: 2 = Potentially Interesting: this paper does not resonate with me, but it might with others in the *ACL community.
Overall Assessment: 2 = Resubmit next cycle: I think this paper needs substantial revisions that can be completed by the next ARR cycle.
Ethical Concerns:
There are no concerns with this submission

Needs Ethics Review: No
Reproducibility: 3 = They could reproduce the results with some difficulty. The settings of parameters are underspecified or subjectively determined, and/or the training/evaluation data are not widely available.
Datasets: 1 = No usable datasets submitted.
Software: 1 = No usable software released.
Knowledge Of Or Educated Guess At Author Identity: No
Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources
Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources
Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources
Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.
Secondary Reviewer: Shangjian Yin (/profile?id=~Shangjian_Yin1)

Add: Official Comment

Response to Reviewer RrUc

Official Comment by Authors 01 Apr 2025, 23:21 (modified: 02 Apr 2025, 12:02)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer RrUc
Revisions (/revisions?id=DevOlHddnd)

Comment:
Thank you for reviewing our paper.

Weaknesses 1: The model size used in the experiment is too small.

- Thank you for your suggestion. In the paper we conducted experiments with the 3B Qwen-LLM, which have already demonstrated that incorporating an LLM as a decoder can significantly enhance SLU performance, even with limited data. We also performed experiments with the 7B model, as shown in the last row of the table. The results indicate that increasing the parameter size of the LLM component can indeed further enhance SLU performance, which aligns with our initial conclusions.

Model	ASR-WER-VP	ASR-WER-VC	NER-F1-VP	SA-F1-VC	SLUE SCORE
GenJSLU (Whisper-small, 244M)	10.58	13.10	59.61	59.14	68.97
Whisper-small + Qwen2.5-3B	14.74	19.01	61.57	63.36	69.35
Whisper-small + Qwen2.5-7B	12.20	16.40	67.79	64.76	72.75

Weaknesses 2: About the NER and Slot Filling

- Slot filling involves extracting content and placing it into predefined slots, while NER focuses on identifying and classifying named entities. These two tasks are not exactly same.

Weaknesses 3: About more convincing baselines

- Our current baseline selection is based on certain results from the SLUE-Benchmark, all of which are official and widely used. Therefore, we believe that choosing these baselines is convincing. We have also considered the

method you mentioned. We do not fully understand your meaning, but if we are not mistaken, you are referring to the approach of sharing the encoder with multiple decoders for different tasks and using different losses for multi-tasking, then this experimental setup is already included in SLUE. If this is not the method you had in mind, could you please provide a more detailed explanation of your suggestion. Thank you.

Weaknesses 4: About the novelty

- Finally, integrating multiple tasks within a unified framework is common in NLU, but in SLU, challenges like the lack of high-quality multi-task aligned data and the limited ability to leverage task interactions persist. A key contribution of our work is the effective use of unaligned data and enhancing multitask interactions to make full use of the task-shared information. Thanks for your useful suggestion.

Add: Official Comment

Reviewer Checklist of Submission3553 by Reviewer RrUc

Edit

Reviewer Checklist by Reviewer RrUc (Peijie Huang (/profile?id=~Peijie_Huang1)) 23 Mar 2025, 22:32

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted

Appropriateness: Yes
Formatting: Yes
Length: Yes
Anonymity: Yes
Limitations: Yes
Responsible Checklist: Yes
Potential Violation Justification: There are no violations with this submission
Need Ethics Review: No
Ethics Review Justification: N/A (I answered no to the previous question)

Add: Official Comment

Official Review of Submission3553 by Reviewer 2XqG

Official Review by Reviewer 2XqG 23 Mar 2025, 00:16 (modified: 28 Mar 2025, 17:27)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 2XqG

Revisions (/revisions?id=Cab2FyQnHR)

Paper Summary:
This paper proposes GenJSLU, a unified generative framework for joint spoken language understanding (SLU) tasks, including automatic speech recognition (ASR), named entity recognition (NER), and sentiment analysis (SA). By introducing a unified representation and a generative framework, GenJSLU enables joint modeling of ASR, NER, and SA, leveraging shared knowledge across tasks and integrating large language models (LLMs) for enhanced generation. Experiments on SLUE-VoxPopuli and SLUE-VoxCeleb datasets demonstrate superior performance compared to benchmark methods.

Summary Of Strengths:

1. Unified Framework: The proposed unified representation for heterogeneous SLU tasks (e.g., combining ASR transcripts with task-specific outputs via control tokens) is a creative solution to leverage multi-source data and task interactions.
2. Effective Joint Modeling: The generative framework successfully integrates ASR, NER, and SA into a single model, enabling cross-task synergy. The dynamic weighted loss balancing ASR and SLU tasks is a practical contribution.

Summary Of Weaknesses:

1. Limited Novelty in Technical Components: While the unified framework is well-motivated, the core technical contributions (e.g., task-specific control tokens, sequence generation with LLMs) resemble existing generative multi-task approaches. The dynamic loss strategy is a straightforward adaptation of length-based weighting.
2. Narrow Model and Dataset Scope: Experiments are limited to Whisper-small and two datasets. Broader validation across diverse architectures (e.g., Wav2Vec 2.0, larger Whisper variants) and multilingual/cross-domain datasets would strengthen claims about generalizability. Furthermore, according to my experience in the NLP field, if one intends to use

autoregressive generative models like GPT for understanding tasks, compared to BERT, significantly more pre-training and downstream tuning data are typically required to achieve comparable performance, which contradicts to the methods introduced in this paper.

Comments Suggestions And Typos:
The ``Limitations" section should not be numbered. In summary, the format and novelty in this paper have significant room for improvement.

Confidence: 4 = Quite sure. I tried to check the important points carefully. It's unlikely, though conceivable, that I missed something that should affect my ratings.
Soundness: 2.5
Excitement: 2.5
Overall Assessment: 2.5 = Borderline Findings
Ethical Concerns:
There are no concerns with this submission

Needs Ethics Review: No
Reproducibility: 2 = They would be hard pressed to reproduce the results: The contribution depends on data that are simply not available outside the author's institution or consortium and/or not enough details are provided.
Datasets: 1 = No usable datasets submitted.
Software: 1 = No usable software released.
Knowledge Of Or Educated Guess At Author Identity: No
Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources
Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources
Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources
Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.

Add: Official Comment

Response to Reviewer 2XqG

Official Comment by Authors 01 Apr 2025, 23:21

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 2XqG

Comment:
Thank you for your thoughtful review of our paper and for recognizing our contribution. We sincerely appreciate your valuable feedback, including your suggestions on our writing format, all of which we have carefully incorporated.

Weaknesses 1: About the novelty

- As you pointed out, our core contribution lies in unifying ASR, NER, and SA through a unified data representation and model architecture. Previous work has generally been limited to SpokenNER and SpokenSA, largely due to the scarcity of aligned datasets for three or more tasks. By proposing a unified structure, we have been able to make more effective use of heterogeneous data, thereby reducing the reliance on extensive multi-task aligned datasets. Moreover, this approach can be extended to a wider range of tasks. Thanks.

Weaknesses 2: About the narrow model and dataset scope

- You raise a key point about data requirements for task understanding. While GPT-like models need more data than BERT, Whisper's pre-trained decoder already possesses a certain level of understanding. Based on this, we fine-tuned it for better task adaptation. Additionally, following your suggestion, we conducted additional experiments on Whisper-Medium. As shown in the last three rows of the table below, the larger model brings significant performance improvements across the three tasks, along with enhanced stability. At the same time, this highlights the importance of our proposed Dynamic Loss method in improving stability for smaller models, particularly in real-world scenarios. Regarding additional languages and cross-domain data, we plan to include them in our future work. Thank you.

Model	ASR-WER-VP	ASR-WER-VC	NER-F1-VP	SA-F1-VC	SLUE SCORE
GenJSLU (Whisper-small, 244M)	10.58	13.10	59.61	59.14	68.97
w/o Dynamic Loss	10.04	12.85	51.24	59.94	66.57
Frozen Encoder	11.13	13.52	47.57	57.24	64.16

Model	ASR-WER-VP	ASR-WER-VC	NER-F1-VP	SA-F1-VC	SLUE SCORE
GenJSLU(Whisper-medium, 769M)	8.77	11.47	69.95	61.60	73.81
w/o Dynamic Loss	8.97	10.93	68.69	61.74	73.49
Frozen Encoder	8.08	11.05	65.18	61.22	72.27

Add: Official Comment

Official Review of Submission3553 by Reviewer hN1p

Official Review by Reviewer hN1p 07 Mar 2025, 23:36 (modified: 28 Mar 2025, 17:27)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer hN1p

Revisions (/revisions?id=StCGvMn9cT)

Paper Summary:
This paper proposes a system for multitask Spoken Language Understanding (SLU) extended with Automatic Speech Recognition (ASR). The main goal of this system is to gain improvement by the means of simultaneous training on multiple SLU datasets and tasks. The prerequisite for this system is that multiple SLU tasks are formulated as a prediction of sequences in a compatible format. An additional benefit of such formulation is that the system training can also include ASR task, which is implemented by inserting the target transcript in front of the SLU ground truth sequence and using this combined sequence as the training target. During inference the system predicts the ASR output first, which serves as an additional context for the SLU sequence generation. After verifying this task formulation separately on the two SLU datasets, SLUE-VoxPopuli (Named Entity Recognition task) and SLUE-VoxCeleb (Sentiment Analysis task), the authors train on both datasets combined and demonstrate the effectiveness of the multitask SLU and ASR combination.

Summary Of Strengths:
1. The proposed method is well motivated, easy to understand and to implement.
2. Evaluations are performed on publicly available widely used benchmarks and utilize open access models.
3. Few ablition studies show the effects of certain design choices in isolation.
4. Some error analysis hints at what kind of inputs would benefit from the proposed method the most.

Summary Of Weaknesses:
1. Some experimental comparisons do not appear informative, because several variables are changed at the same time between trials.
2. Sometimes writing is not precise enough.
3. Some experimental setup details are missing, and it is not clear if the source code is planned to be published.
4. There is no comparison to alternative methods, the only external result is the baseline from the original dataset publication.

Comments Suggestions And Typos:
1. The lines 2 and 3 in Table 2 (SLUE W2V2-L-LL60K and SLUE* Whisper-small) compare two changes at the same time: new formulation of the NER output and new speech encoder. It would be better to add one more result with only one of these changes: SLUE W2V2-L-LL60K with new NER formulation or SLUE Whisper-small with old NER formulation.
2. Similar with lines 2 and 4 in Table 3, the new system introduces LLM decoder and pretraining on the LS960 ASR data at the same time. For the proper comparison, another experiment with GenJSLU including the LS960 ASR pretraining should be reported.
3. It is not clear what conclusion is drawn from the W2V-BERT-2.0 + Qwen-LLM result, this experiment could be removed.
4. GenSLU uses Whisper-small encoder, but what is its decoder, is it also Whisper? What tokenizer is used in GenSLU?
5. What is the length unit in text lines 378--380, is it in tokens?

Typo:"Wave2Vec 2.0" should be "Wav2Vec 2.0".

Confidence: 4 = Quite sure. I tried to check the important points carefully. It's unlikely, though conceivable, that I missed something that should affect my ratings.
Soundness: 3 = Acceptable: This study provides sufficient support for its main claims. Some minor points may need extra support or details.
Excitement: 3 = Interesting: I might mention some points of this paper to others and/or attend its presentation in a conference if there's time.
Overall Assessment: 3 = Findings: I think this paper could be accepted to the Findings of the ACL.
Ethical Concerns:
There are no concerns with this submission

Needs Ethics Review: No
Reproducibility: 3 = They could reproduce the results with some difficulty. The settings of parameters are underspecified or subjectively determined, and/or the training/evaluation data are not widely available.
Datasets: 1 = No usable datasets submitted.
Software: 1 = No usable software released.
Knowledge Of Or Educated Guess At Author Identity: No
Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources
Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources
Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources
Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.

Add: Official Comment

Response to Reviewer hN1p

Official Comment by Authors 01 Apr 2025, 23:21

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer hN1p

Comment:
We deeply appreciate your detailed review and thoughtful feedback, as well as your recognition of our work. And we will continue to refine our language in future versions to ensure greater precision.

Weaknesses 1: Some experimental comparisons do not appear informative

- We have taken your suggestion into account and adjusted the comparisons accordingly: in the revised third row, only the encoder differs from the second row, while in the comparison between the third and fourth rows, only the implementation of NER and SA has been modified. The revised table is as follows. We observe that the special tagging approach for NER performs exceptionally well on Whisper, possibly because it effectively captures NER occurrences as they appear. However, our method performs better in the Joint model, as it avoids conflicts between different formulations of the multi-task approach. As a result, our GenJSLU demonstrates superior overall performance. Thanks.

Model	ASR-WER-VP	ASR-WER-VC	NER-F1-VP	SA-F1-VC	SLUE SCORE
SLUE (W2V2-B-LS960, 96M)	18.40	20.90	49.60	48.60	59.40
SLUE (W2V2-L-LL60K, 317M)	12.10	13.80	50.50	50.10	62.55
SLUE* (Whisper-small, 244M)	10.24	13.20	62.09	51.91	67.42
GenSLU (Whisper-small, 244M)	10.20	12.54	50.29	59.69	66.17

Weaknesses 2: About the experiment details

- To clarify, for GenSLU, both the decoder and tokenizer were based on Whisper-small. Additionally, the length measurement referenced in lines 378–380 of the paper was conducted at the word level rather than the token level. Regarding pretraining Whisper on LS960, we evaluated Whisper-small on the LS960 test set and achieved an impressive WER of 8.99. These results suggest that LS960 is already included in Whisper's extensive pretraining data. Therefore, in our experiments, we did not pretrain GenJSLU on LS960.

Weaknesses 3: About alternative methods and our open-source planning

- We have chosen SLUE as our benchmark because it includes various methods, models, and tasks that fully meet our current needs. Moreover, SLUE is widely used. However, we appreciate your suggestion and will incorporate additional benchmarks and alternative methods in the future. And to support reproducibility, we will open-source all our models and code in the near future. Thanks for your reminding.

Add: Official Comment

Reviewer Checklist of Submission3553 by Reviewer 2XqG

Reviewer Checklist by Reviewer 2XqG 06 Mar 2025, 14:31

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted

Appropriateness: Yes
Formatting: Yes
Length: Yes
Anonymity: Yes
Limitations: Yes
Responsible Checklist: Yes
Potential Violation Justification: There are no violations with this submission
Need Ethics Review: No
Ethics Review Justification: N/A (I answered no to the previous question)

Add: Official Comment

Reviewer Checklist of Submission3553 by Reviewer hN1p

Reviewer Checklist by Reviewer hN1p 04 Mar 2025, 20:40

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted

Appropriateness: Yes
Formatting: Yes
Length: Yes
Anonymity: Yes
Limitations: Yes
Responsible Checklist: Yes
Potential Violation Justification: There are no violations with this submission
Need Ethics Review: No
Ethics Review Justification: N/A (I answered no to the previous question)

Add: Official Comment

Action Editor Checklist by Area Chair ppc7

Action Editor Checklist by Area Chair ppc7 (Thang Vu (/profile?id=~Thang_Vu2)) 03 Mar 2025, 21:33

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted

Appropriateness: Yes
Formatting: Yes
Length: Yes
Anonymity: Yes
Limitations: Yes
Responsible Checklist: Yes
Potential Violation Justification: There are no violations with this submission
Number Of Assignments: Yes
Diversity: Yes
Need Ethics Review: No
Ethics Review Justification: N/A (I answered no to the previous question)
Resubmission: This paper is a not a resubmission
Resubmission Reassignments: This paper is a not a resubmission

Add: Official Comment

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